THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS FULL SYLLABUS PAPER SET 07

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The value of 36^2 is: 1

(a) 1332 **(b)** 72 **(c)** 1260 **(d)** 1296

2. $\sqrt{529}$ is equal to: 1

(a) 23 **(b)** 22 **(c)** 25 **(d)** 24

3. 2×10^{-3} written in usual form is: 1

(a) 0.02 **(b)** 2000 **(c)** 0.002 **(d)** 0.0002

4. The Roman numeral CXLIX stands for: 1

(a) 139 **(b)** 150 **(c)** 159 **(d)** 149

5. One angle of a parallelogram is 72°. Its adjacent angle is: 1

(a) 108° **(b)** 18° **(c)** 216° **(d)** 72°

6. A quadrilateral with only one pair of parallel sides is a: 1

(a) trapezium **(b)** rhombus **(c)** parallelogram **(d)** kite

7. The smallest digit that replaces the blank in 6_17 so that the number is divisible by 3 is: 1

(a) 3 **(b)** 1 **(c)** 6 **(d)** 2

8. The number of terms in the expansion of $(x + y)(p + q)$ is: 1

(a) 6 **(b)** 3 **(c)** 4 **(d)** 2

9. The cost of 12 eggs is ₹ 90. The cost of 18 eggs is: 1

(a) ₹ 120 **(b)** ₹ 135 **(c)** ₹ 150 **(d)** ₹ 108

10. ₹ 500 is divided between A and B in the ratio 2 : 3. A's share is: 1

(a) ₹ 250 **(b)** ₹ 400 **(c)** ₹ 200 **(d)** ₹ 300

11. 0.45 as a percentage is: 1

(a) 0.45% **(b)** 450% **(c)** 45% **(d)** 4.5%

12. The simple interest on ₹ 6000 at 12% per annum for 2 year(s) is: 1

(a) ₹ 2880 **(b)** ₹ 1440 **(c)** ₹ 720 **(d)** ₹ 7440

13. The diagonal of a square of side 12 cm is: 1

(a) 24 cm **(b)** $12\sqrt{2}$ cm **(c)** 12 cm **(d)** $12\sqrt{3}$ cm

14. The legs of a right triangle are 8 cm and 15 cm. Its hypotenuse is: 1

(a) 23 cm **(b)** 16 cm **(c)** 18 cm **(d)** 17 cm

15. The slope of a speed–time reasoning: doubling men halves days. This is: 1

(a) direct proportion **(b)** inverse proportion
(c) equal ratio **(d)** no relation

16. A solid with two parallel congruent polygonal bases joined by rectangles is a: 1

(a) sphere **(b)** cone **(c)** prism **(d)** pyramid

17. The mean of the data 30, 35, 40, 45, 50, 60, 65 is: 1

(a) 44.4286 **(b)** 48.4286 **(c)** 46 **(d)** 46.4286

18. The sum of 47 and its reverse 74 is: 1

(a) 11 **(b)** 121 **(c)** 132 **(d)** 99

19. The area of a rhombus with diagonals 10 cm and 8 cm is: 1

(a) 18 cm² **(b)** 80 cm² **(c)** 40 cm² **(d)** 20 cm²

20. The area of a rectangle of length 18 cm and breadth 11 cm is: 1

(a) 209 cm² **(b)** 29 cm² **(c)** 58 cm² **(d)** 198 cm²

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Express $\frac{1}{64}$ as a power of 2 with a negative exponent. 2

22. Who named 'Hindu numerals' and who spread them in Europe? 2

23. Why is 2 the only even prime? 2

24. Write the ratio 45 minutes : 1 day in simplest form. 2

25. Find the length of the diagonal of a cube face of edge 5 cm. 2

26. The ratio of lengths on a blueprint is 1 : 40. A room 6 m long appears how long on it? 2

27. The mean of 6 numbers is 9. When a seventh is added the mean becomes 10. Find the number. 2

28. A triangle has area 54 cm² and base 12 cm. Find its height. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Write the cubes of the first five natural numbers and state which are odd. 3

30. A parallelogram has adjacent sides 12 cm and 9 cm. Find its perimeter. Is its area 108 cm²? Explain. 3

31. Expand and simplify $(x + 6)(x + 8)$. Verify for $x = 1$. 3

32. At what total amount will ₹ 2400 grow in 4 years at 5% simple interest? 3

33. A cuboid has dimensions 8 cm × 5 cm × 3 cm. Find its total surface area. 3

34. Two numbers add to 30. Find the numbers that give the largest product and the product. 3

35. A ladder of length 20 m rests against a wall with its foot 12 m from the wall. How high up the wall does it reach? 3

OR

A rectangular park measures 5 m by 12 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

36. Find the area of a trapezium with parallel sides 18 cm and 30 cm and distance between them 12 cm. Also find the area of an equal-height rectangle with width 30 cm. 3

OR

A rectangular plot is 18 m by 18 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) Write $\left(\frac{1}{2}\right)^{-5}$ without a negative exponent. 5

(ii) Simplify $\dfrac{(-2)^3 \times (-2)^{-5}}{(-2)^{-1}}$.

OR

(i) Show that $(ab)^n = a^n b^n$ for $a = 2,\ b = 3,\ n = 3$. 5

(ii) Hence find $12^3 \div 6^3$.

38. (i) Define profit, loss and discount. 5

(ii) A fan is marked ₹ 2500 and sold at 12% discount with 18% GST added on the discounted price. Find the final price.

OR

Case study: A television is marked ₹ 30000. A store offers 15% discount, and a festival coupon of ₹ 1500 off. 5

(i) Find the price after the discount.

(ii) Price after the coupon.

(iii) Effective discount percentage.

39. (i) Four numbers have mean 12. If three of them are 8, 10 and 14, find the fourth. 5

(ii) Find the median of the four numbers.

OR

(i) Find the median of 17 numbers where the sorted 9th value is 42. 5

(ii) Why?

40. (i) Prove that parallelograms on the same base between the same parallels have equal area. 5

(ii) Illustrate with an example.

OR

(i) Rectangles of equal perimeter 40 cm can have different areas. Give four examples. 5

(ii) Which is largest?

END OF QUESTION PAPER — **Total: 80 Marks**

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